

HYDROGENATED NBR (HNBR) SYNTHESIS AND LOW-MOONEY PROCESS PATENTS: CATALYSTS, REACTOR MODES, AND DEGRADATION-CONTROL STRATEGIES

1. ABSTRACT

This report synthesizes patent evidence on hydrogenated nitrile rubber (HNBR) preparation, with emphasis on solution hydrogenation of NBR precursors and on process routes that lower molecular weight and Mooney viscosity while preserving elastomeric performance. Across the selected patents, the target material is consistently defined as a nitrile rubber derived from a conjugated diene and an α,β -unsaturated nitrile, subsequently hydrogenated to reduce residual double bonds and improve heat, ozone, oil, and ageing resistance. The landscape splits into two major technical directions: direct hydrogenation of NBR to HNBR using noble-metal catalysts, and tandem or pre-degradation strategies that first reduce polymer molecular weight by metathesis before hydrogenation or, in some cases, combine both transformations in a single reactor concept.

The hydrogenation-focused patents disclose homogeneous catalyst systems based on ruthenium, rhodium, and palladium, often in monochlorobenzene solution, with hydrogen pressures ranging from low single-digit bar in spinning-disk operation to 1200 psi in batch high-pressure reactors and temperatures commonly between 130 and 138 C, or more broadly 60 to 200 C. Several patents emphasize catalyst loading control in the ppm or phr range and the importance of minimizing catalyst residues to avoid discoloration and ageing penalties. Other patents focus on reactor engineering, including spinning disk reactors, to reduce hydrogen pressure requirements and equipment cost.

A second cluster of patents addresses low-molecular-weight HNBR by metathesis-driven chain scission of NBR or by segment interchange between unsaturated polymers. These disclosures use Grubbs-type ruthenium catalysts, Hoveyda/Arlt variants, Wilkinson-type hydrogenation catalysts, and salt additives such as LiBr or CaCl₂ to tune metathesis activity. Reported outcomes include substantial reductions in Mw, narrower polydispersity, and lower Mooney viscosity, with some examples achieving Mooney values near 3 to 40 and Mw reductions from the 200,000-300,000 g/mol range to roughly 50,000-190,000 g/mol. Overall, the evidence supports HNBR synthesis routes that combine controlled unsaturation reduction, catalyst selection, and

process intensification to balance conversion, molecular weight, and final rubber performance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP3827029A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution hydrogenation of at least partially unsaturated nitrile rubber to HNBR using ruthenium, palladium, or rhodium compounds. The emphasis is on maintaining good ageing properties by controlling noble-metal catalyst type and loading, followed by hydrogenation under elevated temperature and pressure without requiring a separate catalyst-removal step. - NBR composition: at least 40% to 90% by weight conjugated diene and at least 10% to 60% by weight alpha,beta-unsaturated nitrile — Claim 1 - Ruthenium amount: 10 ppm to 200 ppm based on at least partially unsaturated nitrile rubber — Claim 1 - Ruthenium amount: 10 ppm to 150 ppm — Claim 1 - Ruthenium amount: 10 ppm to 120 ppm — Claim 1 - Ruthenium amount: 10 ppm to 79 ppm — Claim 1 - Ruthenium amount: 42 ppm to 79 ppm — Claim 1 - Palladium amount: 20 ppm to <67 ppm based on at least partially unsaturated nitrile rubber — Claim 1 - Palladium amount: 44 ppm to <67 ppm — Claim 1 - Rhodium amount: 50 ppm to <270 ppm based on at least partially unsaturated nitrile rubber — Claim 1 - Rhodium amount: 79 ppm to <270 ppm — Claim 1 - Rhodium amount: 50 ppm to 250 ppm — Claim 1 - Rhodium amount: 50 ppm to 170 ppm — Claim 1 - Rhodium amount: 79 ppm to 170 ppm — Claim 1 - Hydrogenation temperature: 60 to 200 C — Claim 1 - Hydrogenation pressure: 700000 pascals to 15000000 pascals — Claim 1 - Hydrogenation time: 1 to 24 hours — Claim 1

Relevant Experimental Evidence - No worked-example sections were present in the patent text; report evidence falls back to claims/description. - Detailed Description: Description Method for producing hydrogenated nitrile rubber and HNBR compositions thereof The present invention relates to a method for producing hydrogenated nitrile rubber (HNBR) having good ageing properties in the presence of ruthenium

Technical Relevance This patent is directly relevant because it defines a practical HNBR hydrogenation window using Ru/Pd/Rh catalysts and explicitly addresses ageing performance, catalyst residue control, and industrially relevant pressure/temperature ranges. It is a core direct-synthesis disclosure for HNBR.

Patent 2

- Patent Number: US7919563B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent prepares low molecular weight HNBR by first subjecting NBR to metathesis in the presence of a ruthenium or osmium catalyst and optionally a co-olefin, then hydrogenating the metathesis product. The process is designed to reduce Mw and polydispersity while retaining the HNBR character of the polymer. - Target HNBR Mw: 30000 to 250000 — Claim 1 - Target HNBR Mooney viscosity: 3 to 50 — Claim 1 - Target HNBR polydispersity index: less than 2.5 — Claim 1 - Conjugated diene content: butadiene/isoprene class, with examples using butadiene-based NBR — Description - Alpha,beta-unsaturated nitrile content: acrylonitrile class — Description - Cement concentration: 6 or 15% by weight — Examples 1-4 - Agitator speed: 600 rpm — Examples 1-4 - Hydrogen pressure: 1200 psi — hydrogenation conditions - Hydrogenation catalyst loading: 0.08 phr Wilkinson's catalyst — hydrogenation conditions - Triphenylphosphine: 1 phr — hydrogenation conditions - Hydrogenation reactor temperature: 138 C — hydrogenation conditions - Degassing: 3 times with H₂ at 100 psi — hydrogenation procedure - Metathesis reactor temperature: 80 C — Table 1 - Metathesis catalyst loading: 0.05 phr, 0.10 phr, or 0.25 phr — Table 1 - Co-olefin pressure/concentration: 500 psi ethylene or 150 g 1-hexene — Table 1 - Reaction time: 240 min — hydrogenation profile and Table 1 - Starting polymer acrylonitrile content: 34 mol % — Examples - Starting polymer Mooney viscosity: 35 — Examples - Starting polymer Mw: 296000 g/mol — Examples - Starting polymer Mn: 85000 g/mol — Examples - Product Mn: 27000 g/mol — typical product statement - Product Mw: 54000 g/mol — typical product statement - PDI: 2.0 — typical product statement

Relevant Experimental Evidence - Example 1: Ethylene-assisted metathesis at 15% cement and 80 C with 0.05 phr catalyst gave HNBR with reduced Mw and lower Mooney viscosity. - Example 2: Increasing catalyst loading to 0.10 phr at the same cement and temperature further reduced molecular weight and viscosity. - Example 3: Lower cement concentration with 0.25 phr catalyst produced the lowest PDI reported in the table. - Example 4: 1-Hexene as co-olefin at 80 C also produced low-Mw HNBR, demonstrating co-olefin flexibility.

Technical Relevance This patent is highly relevant because it couples metathesis chain scission with hydrogenation to make low-Mooney HNBR. It provides concrete process windows for catalyst loading, co-olefin choice, and hydrogenation conditions.

Patent 3

- Patent Number: US9079979B2
- Patent Title: Process for the preparation of hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses simultaneous hydrogenation and metathesis of NBR in the presence of hydrogen and a ruthenium- or osmium-based catalyst, optionally with a co-olefin. The process is intended to generate low molecular weight HNBR in a single step, using catalysts capable of both metathesis and hydrogenation. - Target HNBR Mw: 20000 to 250000 — Description - Target HNBR Mooney viscosity: 1 to 50 — Description - Target HNBR polydispersity index: less than 2.8 — Description - NBR conjugated diene: C4-C6 dienes, especially butadiene/isoprene — Description - Alpha,beta-unsaturated nitrile: C3-C5 nitriles, especially acrylonitrile — Description - Optional copolymerizable monomers: 0 to 40% by weight — Description - NBR feed Mooney viscosity: 24 to 70, preferably 28 to 40 — Description - Hydrogen pressure in batch hydrogenation context: 65 to 90 bar — Description - Catalyst amount in prior art example cited: 307 ppm and 61 ppm Ru — Description - Catalyst amount in prior art example cited: 0.5 phr Grubbs I catalyst — Description - Hydrogenation/metathesis reaction temperature: 130 C to 138 C — Examples 1-2 - Hydrogen pressure: 7 bar for degassing — Examples 1-2 - Reaction pressure: 85 bar — Examples 1-2 - Reaction time: 240 min — Examples 1-2 - Catalyst amount: 0.057 phr, 0.080 phr, 0.060 phr, 0.055 phr — Table 1 - Metal content: 65.6 ppm, 91.3 ppm, 65.6 ppm, 65.6 ppm — Table 1 - Residual double bond at 60 min: 2.0%, 0.7%, 15.4%, 2.9% — Table 2 - Residual double bond at 120 min: 0.4%, 0.5%, 3.1%, 0.6% — Table 2 - Residual double bond at 180 min: 0.3%, 0.3%, 1.3%, 0.3% — Table 2 - Residual double bond at 240 min: 0.6% for HNBR 3 — Table 2 - Product Mw: 191541, 135772, 195304, 144212 g/mol — Table 3 - Product Mn: 68643, 58259, 64130, 59851 g/mol — Table 3 - Product PDI: 2.79, 2.33, 3.05, 2.45 — Table 3 - Product Mooney viscosity: 38.2, 20.2, 69.3, 32.6 — Table 3

Relevant Experimental Evidence - Example 1: Grubbs III catalyst gave faster hydrogenation than Wilkinson's or Grubbs II at equal metal content. - Example 1: Lower catalyst loading still achieved substantial RDB reduction and lower Mw HNBR. - Example 1: HNBR 2 showed the lowest Mooney viscosity among the four samples. - Example 1: Comparative catalysts produced slower hydrogenation profiles and higher Mooney viscosity.

Technical Relevance This patent is relevant because it demonstrates a one-pot route to low-Mw HNBR by simultaneous metathesis and hydrogenation. It also provides comparative catalyst data showing the benefit of Grubbs III-type systems for faster conversion.

Patent 4

- Patent Number: US7666950B2
- Patent Title: Process for preparing hydrogenated nitrile rubbers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent prepares hydrogenated nitrile rubber in a spinning disk reactor, using hydrogenation under comparatively low hydrogen pressures relative to conventional batch processes. The disclosure emphasizes homogeneous hydrogenation catalysts, especially rhodium- or ruthenium-containing systems, to reduce equipment cost and reaction severity. - Hydrogen pressure: 3.5 bar to 100 bar — Claim 1 - Hydrogenation catalyst: homogeneous hydrogenation catalyst — Claim 4 - Catalyst family: rhodium, ruthenium, titanium, platinum, iridium, palladium, rhenium, osmium, cobalt or copper — Claim 5 - Preferred catalyst family: rhodium or ruthenium containing catalyst — Claim 6 - NBR feed Mooney viscosity: 30 to 90 — Description - NBR feed Mooney viscosity: 30 to 70 — Description - NBR feed Mooney viscosity: 30 to 50 — Description - NBR feed Mw: 200000 to 500000 — Description - NBR feed Mn: 80000 to 150000 — Description - NBR feed PDI: 2.0 to 6.0 — Description - Hydrogenation catalyst loading: 0.36 phr, 0.36 phr, 0.60 phr — Examples 1-3 - Hydrogen pressure: 10 bar, 3.5 bar, 9.8 bar — Table 1 - Disk temperature: 135 C, 135 C, 150 C — Table 1 - Reaction time: 240 min — Table 1 - Cement concentration: 15%, 15%, 6% — Table 1 - Co-olefin: ethylene or 1-hexene — Examples 1-4 - Co-olefin concentration: 500 psi or 150 g — Table 1 - Agitator speed: 600 rpm — Examples - Solvent: monochlorobenzene — Examples - Starting polymer acrylonitrile content: 34% by weight — Examples - Starting polymer Mooney viscosity: 28.2 — Examples - Starting polymer Mw: 255000 g/mol — Examples - Starting polymer Mn: 77000 g/mol — Examples - Product Mw: 143000, 139000, 161000 g/mol — Table 2 - Product Mn: 69000, 62000, 81000 g/mol — Table 2 - Product PDI: 2.09, 2.23, 1.99 — Table 2 - Product solution viscosity: 90.5, 40.5, 91.0 cP — Table 2 - Product RDB: 6.0%, 25.4%, 0.9% — Table 2

Relevant Experimental Evidence - Example 1: 10 bar hydrogen and 0.36 phr catalyst produced HNBR with 6.0% RDB and Mw 143000 g/mol. - Example 2: Lower hydrogen pressure of 3.5 bar still gave substantial hydrogenation but higher residual unsaturation. - Example 3: Higher catalyst loading and 150 C produced the lowest RDB of 0.9% and PDI of 1.99. - Example 4: 1-Hexene co-olefin was also compatible with the spinning disk reactor process.

Technical Relevance This patent is relevant because it offers a reactor-intensified HNBR hydrogenation process that lowers pressure requirements. It is useful for scale-up concepts where heat and mass transfer are critical.

Patent 5

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a two-step route in which NBR is first subjected to olefin metathesis with a ruthenium or osmium carbene catalyst and then hydrogenated to form low molecular weight HNBR. The process uses a co-olefin and a homogeneous catalyst system to reduce Mw and Mooney viscosity while preserving HNBR functionality. - Target HNBR Mw: 30000 to 250000 — Claim 1 - Target HNBR Mooney viscosity: less than 10 — Claim 1 - Target HNBR polydispersity index: less than 2.5 — Claim 1 - NBR feed acrylonitrile content: 34% by weight — Examples - NBR feed Mooney viscosity: 28 Mooney units — Examples - NBR feed Mw: 214000 g/mol — Examples - NBR feed Mn: 67000 g/mol — Examples - NBR feed PDI: 3.2 — Examples - NBR solution concentration: 12% — Examples - NBR mass: 518 g — Examples - Monochlorobenzene mass: 4300 g — Examples - Degassing: 3 times with H₂ at 100 psi — Examples - Reactor agitation: 600 rpm — Examples - Initial reactor temperature: 130 C — Examples - Hydrogenation pressure: 85 bar — Examples - Hydrogenation temperature: 138 C — Examples - Reaction time: 240 min — Examples - Catalyst amount: 0.041 phr, 0.045 phr, 0.060 phr — Table 1 - Metal content: 65.6 ppm — Table 1 - Co-catalyst amount: 1.0 phr triphenylphosphine — Table 1 - Residual double bond at 60 min: 2.5%, 7.0%, 15.4% — Table 2 - Residual double bond at 120 min: 0.6%, 4.5%, 3.1% — Table 2 - Residual double bond at 180 min: 0.5%, 3.9%, 1.3% — Table 2 - Residual double bond at 240 min: 0.4%, 3.5%, 0.7% — Table 2 - Product Mw: 113371, 133758, 195304 g/mol — Table 3 - Product Mn: 48352, 54595, 64130 g/mol — Table 3 - Product PDI: 2.34, 2.45, 3.05 — Table 3 - Product Mooney viscosity: 33.3, 41.3, 69.3 — Table 3

Relevant Experimental Evidence - Example 1: Hoveyda catalyst gave the fastest hydrogenation profile and the lowest Mooney viscosity among the three catalysts. - Example 1: Arlt catalyst also reduced Mw and Mooney viscosity relative to the starting NBR. - Example 1: Wilkinson's catalyst required triphenylphosphine co-catalyst and gave the slowest hydrogenation profile. - Example 1: The invention catalysts outperformed the industrial norm in hydrogenation efficiency.

Technical Relevance This patent is directly relevant because it combines metathesis-driven molecular-weight reduction with hydrogenation to make low-Mooney HNBR. It provides a clear

catalyst comparison and process window for producing lower-viscosity HNBR.

Patent 6

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses hydrogenated nitrile rubber with low molecular weight and low Mooney viscosity, prepared by metathesis of NBR in the presence of a ruthenium catalyst and co-olefin, followed by hydrogenation. The examples show both ethylene and 1-hexene as co-olefins and demonstrate substantial reductions in Mw and PDI. - Target HNBR Mw: 30000 to 250000 — Claim 1 - Target HNBR Mooney viscosity: less than 10 — Claim 1 - Target HNBR polydispersity index: less than 2.5 — Claim 1 - Target HNBR Mw: 40000 to 220000 — Claim 2 - Target HNBR polydispersity index: less than 2.3 — Claim 3 - Starting polymer acrylonitrile content: 34 mol % — Examples - Starting polymer Mooney viscosity: 35 — Examples - Starting polymer Mw: 296000 g/mol — Examples - Starting polymer Mn: 85000 g/mol — Examples - Cement concentration: 6 or 15% by weight — Examples - Agitator speed: 600 rpm — Examples - Hydrogen pressure: 1200 psi — hydrogenation conditions - Hydrogenation catalyst loading: 0.08 phr Wilkinson's catalyst — hydrogenation conditions - Triphenylphosphine: 1 phr — hydrogenation conditions - Hydrogenation reactor temperature: 138 C — hydrogenation conditions - Degassing: 3 times with H₂ at 100 psi — hydrogenation procedure - Metathesis reactor temperature: 80 C — Table 1 - Metathesis catalyst loading: 0.05 phr, 0.10 phr, 0.25 phr — Table 1 - Co-olefin: ethylene or 1-hexene — Examples 1-4 - Co-olefin concentration: 500 psi or 150 g — Table 1 - Reaction time: 240 min — Table 1 - Product Mn: 27000 g/mol — typical product statement - Product Mw: 54000 g/mol — typical product statement - Product PDI: 2.0 — typical product statement - Experiment 1 Mn: 73000 g/mol — Table 2 - Experiment 1 Mw: 189000 g/mol — Table 2 - Experiment 1 PDI: 2.59 — Table 2 - Experiment 1 Mooney viscosity: 43 — Table 2 - Experiment 2 Mn: 60000 g/mol — Table 2 - Experiment 2 Mw: 136000 g/mol — Table 2 - Experiment 2 PDI: 2.27 — Table 2 - Experiment 2 Mooney viscosity: 28 — Table 2 - Experiment 3 Mn: 31000 g/mol — Table 2 - Experiment 3 Mw: 59000 g/mol — Table 2 - Experiment 3 PDI: 1.90 — Table 2 - Experiment 3 Mooney viscosity: 3 — Table 2 - Experiment 4 Mn: 55000 g/mol — Table 2 - Experiment 4 Mw: 111000 g/mol — Table 2 - Experiment 4 PDI: 2.02 — Table 2 - Experiment 4 Mooney viscosity: 31 — Table 2

Relevant Experimental Evidence - Example 1: Ethylene at 500 psi and 0.05 phr catalyst gave a product with Mw 189000 g/mol and Mooney 43. - Example 2: Higher catalyst loading at the same conditions reduced Mw further to 136000 g/mol and Mooney to 28. - Example 3: Lower cement concentration and 0.25 phr catalyst produced the lowest PDI and Mooney viscosity of 3. - Example

4: 1-Hexene co-olefin also produced low-Mw HNBR, confirming co-olefin versatility.

Technical Relevance This patent is relevant because it is a direct low-Mooney HNBR process patent with explicit process and property data. It shows how metathesis conditions can be tuned to achieve very low viscosity HNBR suitable for shaped articles.

Patent 7

- Patent Number: EP2519350B1
- Patent Title: Highly active metathesis catalysis selective for romp and rcm
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is primarily a metathesis-catalyst patent, but it explicitly discloses use of the novel ruthenium complexes for depolymerization of NBR and for hydrogenation of rubbers containing carbon-carbon double bonds. The HNBR relevance lies in the catalyst platform and its stated applicability to depolymerized HNBR and rubber hydrogenation, rather than in a full HNBR manufacturing example set. - Ligand preparation scale: 5.0 mmol — Examples 1-19 - Ru complex preparation scale: 1.0 mmol — Examples 1-19 - Ligand synthesis temperature: 85 C — Example 1 - Ligand synthesis time: overnight — Example 1 - Ru complex formation temperature: 20 to 25 C — Example 1 - Ru complex formation time: 30 to 60 min — Example 1 - Ligand 3a yield: 86% — Example 1 - Ru complex 4a yield: 4% — Example 1 - Ligand 3b yield: 91% — Example 2 - Ru complex 4b yield: 29% — Example 2 - Ligand 3c yield: 54% — Example 3 - Ru complex 4c yield: 5% — Example 3 - Ligand 3d yield: 62% — Example 4 - Ru complex 4d yield: 32% — Example 4 - Ru complex 4e yield: 35% — Example 5 - Ligand 3f yield: 63% — Example 6 - Ru complex 4f yield: 14% — Example 6 - Ligand 3g yield: 56% — Example 7 - Ru complex 4g yield: 8% — Example 7 - Ligand 3h yield: 32% — Example 8 - Ru complex 4h yield: 42% — Example 8 - Ligand 3j yield: 93% — Example 9 - Ru complex 4j yield: 48% — Example 9 - Ligand 3k yield: 83% — Example 10 - Ru complex 4k yield: 68% — Example 10 - Ligand 3m yield: 49% — Example 11 - Ru complex 4m yield: 52% — Example 11 - Ligand 3n yield: 45% — Example 12 - Ru complex 4n yield: 64% — Example 12 - Ligand 3p yield: 67% — Example 13 - Ru complex 4p yield: 26% — Example 13 - Ligand 3q yield: 46% — Example 14 - Ru complex 4q yield: 38% — Example 14 - Ligand 3r yield: 44% — Example 15 - Ru complex 4r yield: 14% — Example 15 - Ligand 3s yield: 51% — Example 16 - Ru complex 4s yield: 85% — Example 16 - Ligand 3t yield: 38% — Example 17 - Ru complex 4t yield: 23% — Example 17 - Ligand 3u yield: 28% — Example 18 - Ru complex 4u yield: 32% — Example 18 - Ligand 3v yield: 81% — Example 19 - Ru complex 4v yield: 10% — Example 19 - NBR depolymerization temperature: 30 to 100 C — claims/description - NBR depolymerization with mixed catalysts: 30 to 100 C — description

Relevant Experimental Evidence - Example 1: Ru complex 4a was synthesized from ligand 3a and characterized by NMR. - Example 2: Ru complex 4b was obtained in 29% yield from ligand 3b. - Example 7: Ligand 3g was confirmed by LC-MS before Ru complex formation. - Example 19: Ru complex 4v was obtained and characterized, expanding the catalyst library.

Technical Relevance This patent is relevant because it supplies a family of ruthenium metathesis catalysts explicitly intended for NBR depolymerization and hydrogenation-related rubber modification. It supports catalyst design for HNBR precursor degradation and segment interchange chemistry.

Patent 8

- Patent Number: US8957161B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses metathetic degradation of NBR using ruthenium metathesis catalysts in the presence of salt additives, especially alkaline-earth metal chlorides, to increase catalyst activity. The process is carried out in monochlorobenzene at room temperature with 1-hexene as co-olefin, and the resulting lower-molecular-weight nitrile rubber can serve as an HNBR precursor after hydrogenation. - Salt:catalyst weight ratio: 0.01:1 to 10000:1 — Claim 1 - Preferred salt: calcium chloride or magnesium chloride — Claim 2 - Catalyst class: complex catalyst based on molybdenum, tungsten, osmium or ruthenium — Claim 3 - NBR feed acrylonitrile content: 35% by weight — Examples - NBR feed Mooney viscosity: 34 Mooney units — Examples - NBR feed Mw: 240000 g/mol — Examples - NBR feed Mn: 100000 g/mol — Examples - NBR feed PDI: 2.4 — Examples - NBR mass: 40 g — Examples - Chlorobenzene mass: 293.3 g — Examples - NBR dissolution time: 10 h — Examples - 1-hexene amount: 0.8 g (2 phr) — Examples - Reaction temperature: room temperature — Examples - Catalyst solution solvent: 20 g MCB — Examples - Reaction quench: ethyl vinyl ether about 0.2 ml — Examples - DMAc dilution: 5 ml — Examples - GPC flow rate: 0.5 ml/min — Examples - Catalyst amount: 0.05 phr Grubbs II — Table 1 - Ru amount: 60 ppm — Table 1 - Salt amount: 2 phr LiBr, LiCl, LiI — Table 1 - CaCl₂ amounts: 0.01, 0.05, 0.1, 0.5, 1, 2, 5, 10 phr — Table 1 - Reaction time points: 0, 30, 60, 90, 180, 420 min — Tables - Mw at 0 min: 186832 g/mol — Table 1.01 - Mw at 420 min without salt: 53852 g/mol — Table 1.01 - Mw at 420 min with 2 phr LiBr: 51827 g/mol — Table 1.02 - Mw at 420 min with 2 phr LiCl: 21337 g/mol — Table 1.03 - Mw at 420 min with 2 phr LiI: 51827 g/mol — Table 1.04 - PDI at 420 min without salt: 1.8 — Table 1.01 - PDI at 420 min with 2 phr LiCl: 1.6 — Table 1.03

Relevant Experimental Evidence - Example 1.01: Grubbs II without salt reduced Mw from 186832 to 53852 g/mol over 420 min. - Example 1.02: LiBr accelerated early degradation but did not outperform LiCl at final Mw. - Example 1.03: LiCl gave the strongest Mw reduction, reaching 21337 g/mol at 420 min. - Example 1.04: LiI also accelerated degradation but was less effective than LiCl. - Example 1.05 and beyond: CaCl₂ was explored over a wide phr range as the preferred additive.

Technical Relevance This patent is relevant because it improves metathesis-based NBR chain scission, a key precursor step for low-Mw HNBR. The salt-additive strategy can lower catalyst demand and improve degradation efficiency before hydrogenation.

Patent 9

- Patent Number: US7737233B2
- Patent Title: Catalyst systems and their use for metathesis reactions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses catalyst systems for metathesis of nitrile rubber using a metathesis catalyst together with alkaline-earth metal chloride additives, especially calcium chloride. The examples show that salt addition can markedly change degradation rate and final molecular weight, making the process relevant to preparing low-Mw NBR precursors for HNBR synthesis. - Catalyst system salt:catalyst ratio: 0.01:1 to 10000:1 — Claim 1 - Preferred salt: calcium chloride or magnesium chloride — Claim 2 - Catalyst class: molybdenum, tungsten, osmium or ruthenium complex catalyst — Claim 3 - NBR feed acrylonitrile content: 34% by weight — Examples - NBR feed Mooney viscosity: 31 Mooney units — Examples - NBR feed Mw: 186.832 g/mol — Examples - NBR feed Mn: 60000 g/mol — Examples - NBR feed PDI: 3.1 — Examples - NBR mass: 40 g — Examples - Chlorobenzene mass: 293.3 g — Examples - 1-hexene amount: 0.8 g (2 phr) — Examples - Reaction temperature: room temperature — Examples - Catalyst solution solvent: 20 g MCB — Examples - Catalyst amount: 20 mg Grubbs II catalyst, 0.05 phr, 60 ppm Ru — Table 1.01 - Salt amount: 2 phr LiBr, LiCl, or LiI — Table 1.02-1.04 - CaCl₂ amounts: 0.01, 0.05, 0.1, 0.5, 1, 2, 5, 10 phr — Table 1.05-1.12 - Reaction time points: 0, 30, 60, 90, 180, 420 min — Tables - Mw at 420 min without salt: 53852 g/mol — Table 1.01 - Mw at 420 min with LiBr: 51827 g/mol — Table 1.02 - Mw at 420 min with LiCl: 21337 g/mol — Table 1.03 - Mw at 420 min with LiI: 51827 g/mol — Table 1.04 - PDI at 420 min without salt: 1.8 — Table 1.01 - PDI at 420 min with LiCl: 1.6 — Table 1.03

Relevant Experimental Evidence - Comparative example 1.01: Grubbs II alone reduced Mw steadily but not as strongly as the best salt-assisted cases. - Comparative example 1.02: LiBr

improved early kinetics but final Mw remained relatively high. - Comparative example 1.03: LiCl gave the lowest final Mw and PDI among the lithium salts. - Example 1.05 onward: CaCl₂ was screened across 0.01 to 10 phr to identify an optimal additive window.

Technical Relevance This patent is relevant because it shows how salt additives can tune NBR metathesis, which is a precursor route to low-Mw HNBR. It complements hydrogenation patents by improving the upstream chain-shortening step.

Patent 10

- Patent Number: US8557921B2
- Patent Title: Production of meta-block copolymers by polymer segment interchange
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent describes metathesis-driven segment interchange between two or more ethylenically unsaturated polymers to form meta-block copolymers, and it explicitly includes partially hydrogenated nitrile/butadiene rubber as one reagent polymer. The HNBR relevance is indirect but important: partially hydrogenated NBR can participate in metathesis-based block formation and rubber modification under nitrogen with Grubbs II catalyst or tungsten-based systems. - Grubbs II catalyst identity: benzylidene[1,3-bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro(tricyclohexyl phosphine)ruthenium — Examples - Inert atmosphere: nitrogen gas — Examples - PBD approximate Mw: 250000 g/mol — Unsaturated reagent polymers - PBD ethylenic carbon content: 42 percent — Unsaturated reagent polymers - HPBD hydrogenation level: 90 percent — Unsaturated reagent polymers - HPBD unsaturation level: 5.5 mole percent — Unsaturated reagent polymers - HPBD Tm: 109 C — Unsaturated reagent polymers - HNBR acrylonitrile content: 33 percent — Unsaturated reagent polymers - HNBR Mooney viscosity: 61 — Unsaturated reagent polymers - HNBR hydrogenation of ethylenic unsaturation: approximately 90 percent — Unsaturated reagent polymers - HNBR unsaturation level: 0.81 mole percent — Unsaturated reagent polymers - HNBR Tg: -25 C — Unsaturated reagent polymers - Reactor volume: 2 L Parr batch reactor — Butadiene copolymer polymerizations - Polymerization pressure: 3.4 MPa — Butadiene copolymer polymerizations - Polymerization temperature: not explicitly stated for the reactor charge, but reactor is heated to polymerization temperature — Butadiene copolymer polymerizations - Example 1 catalyst amount: 8 mg Grubbs II — Example 1 - Example 1 reaction time: 1 hour — Example 1 - Example 1 polymer product recovered: 0.45 g — Example 1 - Example 2 catalyst amount: 9 mg Grubbs II — Example 2 - Example 2 reaction temperature: 105 C — Example 2 - Example 2 reaction time: 1 hour — Example 2 - Example 2 product recovered: 0.47 g — Example 2 - Example 3 catalyst amount: 9 mg Grubbs II — Example 3 - Example 3 reaction temperature: 105 C — Example 3 - Example 3 reaction time: 1 hour —

Example 3 - Example 3 product recovered: 1.37 g — Example 3 - Example 4 catalyst amount: 9 mg Grubbs II — Example 4 - Example 4 reaction temperature: 105 C — Example 4 - Example 4 reaction time: 1 hour — Example 4 - Example 4 product recovered: 0.47 g — Example 4 - Example 5 catalyst amount: 9 mg Grubbs II — Example 5 - Example 5 reaction temperature: 105 C — Example 5 - Example 5 reaction time: 1 hour — Example 5 - Example 5 product recovered: 0.47 g — Example 5 - Example 6 catalyst amount: 9 mg Grubbs II — Example 6 - Example 6 reaction temperature: 105 C — Example 6 - Example 6 reaction time: 1 hour — Example 6 - Example 6 product recovered: 0.47 g — Example 6 - Example 7 tungsten hexachloride catalyst: 8000 ppm — Example 7 - Example 7 tri-n-butylmethyltin co-catalyst: 13000 ppm — Example 7 - Example 7 reaction time: 1 hour — Example 7 - Example 7 product recovered: 0.47 g — Example 7 - Example 8 catalyst amount: 9 mg Grubbs II — Example 8 - Example 8 reaction temperature: 105 C — Example 8 - Example 8 reaction time: 1 hour — Example 8 - Example 8 product recovered: 0.47 g — Example 8 - Example 9 tungsten hexachloride catalyst: 8000 ppm — Example 9 - Example 9 tri-n-butylmethyltin co-catalyst: 13000 ppm — Example 9 - Example 9 reaction time: 1 hour — Example 9 - Example 9 product recovered: 0.47 g — Example 9 - Example 10 catalyst amount: 9 mg Grubbs II — Example 10 - Example 10 reaction temperature: 105 C — Example 10 - Example 10 reaction time: 1 hour — Example 10 - Example 10 product recovered: 0.71 g — Example 10

Relevant Experimental Evidence - Example 1: HPBD and HNBR were converted into a meta-block copolymer using 8 mg Grubbs II catalyst. - Example 2: Unsaturated polyurethane and poly(ethylene-co-butadiene) formed a segment-interchange product after 1 hour. - Example 7: Tungsten hexachloride and tri-n-butylmethyltin enabled segment interchange between two ethylene/butadiene copolymers. - Example 10: A ternary blend including HNBR-related unsaturated polymers produced a meta-block copolymer with ordered microphase morphology.

Technical Relevance This patent is relevant because it explicitly includes partially hydrogenated nitrile/butadiene rubber as a reagent polymer in metathesis-based block copolymer formation. It broadens the HNBR-related synthesis landscape beyond simple hydrogenation to post-hydrogenation polymer architecture control.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Direct hydrogenation patents (EP3827029A1, US9079979B2, US7666950B2) focus on Ru/Rh/Pd catalyst systems, with EP3827029A1 emphasizing ppm-level noble-metal control and ageing stability, US9079979B2 emphasizing simultaneous hydrogenation/metathesis, and US7666950B2 emphasizing spinning-disk reactor intensification.
- Low-Mw HNBR patents (US7919563B2, EP2473281B1, US7772328B2) all use metathesis-assisted chain scission before or during hydrogenation; US7919563B2 and US7772328B2 use Grubbs/Wilkinson-type systems with co-olefins, while EP2473281B1 compares Hoveyda, Arlt, and Wilkinson catalysts.

- Reaction media are consistently aromatic chlorinated solvents, especially monochlorobenzene/chlorobenzene, across the hydrogenation and metathesis patents; solution concentrations commonly range from 4% to 15% solids or 6% to 15% cement.
- Hydrogenation conditions vary widely: EP3827029A1 claims 60 to 200 C and 700000 to 15000000 Pa, US9079979B2 uses 130 to 138 C and 85 bar, and US7666950B2 demonstrates 3.5 to 10 bar in a spinning disk reactor with 138 C hydrogenation.
- Catalyst loadings are generally low in direct hydrogenation patents, often in phr or ppm ranges, while metathesis/degradation patents use 0.041 to 0.25 phr catalyst or 8000 to 13000 ppm catalyst/co-catalyst systems.
- Co-olefin use is a recurring lever in the metathesis-based patents: ethylene and 1-hexene are the most common, with 1-hexene often used at 2 phr or 150 g scale to promote chain scission.
- Salt additives are a distinct trend in the degradation patents US8957161B2 and US7737233B2, where LiBr, LiCl, LiI, and CaCl₂ are used to modulate metathesis activity and final Mw; this strategy is absent from the direct hydrogenation patents.
- Property outcomes differ by route: direct hydrogenation patents mainly target ageing stability and catalyst residue control, whereas metathesis-assisted patents explicitly report Mw reductions to roughly 54000 to 195000 g/mol and Mooney viscosity reductions down to 3 to 41.
- US8557921B2 is the most structurally different disclosure, using HNBR as a reagent in meta-block copolymer formation rather than as the final product; it nevertheless confirms HNBR compatibility with metathesis chemistry.

4. CONCLUSION

The evidence supports two practical HNBR synthesis paradigms: direct solution hydrogenation of NBR with noble-metal catalysts, and metathesis-assisted molecular-weight reduction followed by hydrogenation. For direct HNBR manufacture, the strongest evidence favors Ru/Rh/Pd catalyst systems in chlorinated aromatic solvents, with hydrogenation at roughly 60 to 200 C and elevated pressure, while careful control of catalyst residue is important for ageing performance. For low-Mooney HNBR, the most effective route is metathesis chain scission of NBR using Grubbs-type ruthenium catalysts, optionally with ethylene or 1-hexene co-olefins, followed by hydrogenation; salt additives such as CaCl₂ or LiCl can further accelerate degradation and lower final Mw. Based on the disclosed data, a robust synthesis strategy would combine a low-residue hydrogenation catalyst, monochlorobenzene solvent, controlled solids loading around 4% to 15%, and either a spinning-disk or high-pressure solution reactor depending on throughput and pressure constraints. For lowest viscosity products, the metathesis-first route with optimized catalyst loading and salt assistance appears most effective.

5. REFERENCES

- EP3827029A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3827029A1/en>

- US7919563B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7919563B2/en>
- US9079979B2 | Process for the preparation of hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9079979B2/en>
- US7666950B2 | Process for preparing hydrogenated nitrile rubbers | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7666950B2/en>
- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>
- US7772328B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7772328B2/en>
- EP2519350B1 | Highly active metathesis catalysis selective for romp and rcm | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP2519350B1/en>
- US8957161B2 | Process for reacting chemical compounds in the presence of catalyst systems | Not disclosed | US | 2015 | <https://patents.google.com/patent/US8957161B2/en>
- US7737233B2 | Catalyst systems and their use for metathesis reactions | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7737233B2/en>
- US8557921B2 | Production of meta-block copolymers by polymer segment interchange | Not disclosed | US | 2013 | <https://patents.google.com/patent/US8557921B2/en>